TEXT CLEANING BASICS

Task 26 | Data Analytics Internship

Veda Technology

1. Objective

The objective of this task is to standardize text fields in a customer dataset by fixing inconsistent spacing, capitalization, and labels. Cleaning text helps make records more consistent and easier to search, group, and analyse.

2. Tools and Dataset

Tools: Python (pandas) and Microsoft Excel
Dataset: Synthetic customer dataset created for practice
Records: 20 fictional customer rows

3. Cleaning Steps

Field / issue	Cleaning method	Example
Extra spaces	Trim leading/trailing spaces and collapse repeated spaces.	' aNita sharma ' → 'Anita Sharma'
Capitalization	Use title case for names and city names.	'BENGALURU' → 'Bengaluru'
Email formatting	Remove surrounding spaces and convert to lowercase.	' ANITA@EXAMPLE.COM ' → 'anita@example.com'
Inconsistent labels	Standardize customer segment values.	'consumer' → 'Consumer'
Country text	Trim spaces and standardize capitalization.	' India ' → 'India'

4. Python Functions Used

strip() removes spaces from the beginning and end of a string.
split() and join() help collapse repeated spaces into a single space.
title() converts text to title case for readability.
lower() converts email addresses to lowercase.
pandas applies these transformations across entire columns.

5. Before-and-After Examples

Field	Before	After
Customer Name	aNita sharma	Anita Sharma
Email	ANITA.SHARMA@EXAMPLE.COM	anita.sharma@example.com
City	BENGALURU	Bengaluru
Customer Segment	consumer	Consumer
Country	India	India

6. Results

The Python script was run successfully on the included 20-row synthetic dataset and produced a cleaned CSV file. The Excel workbook contains separate sheets for the original data, cleaned data, before-and-after examples, and a cleaning summary.

7. Why Text Cleaning Matters

Inconsistent spacing, capitalization, and category labels can cause duplicate-looking values and make grouping or searching less reliable. Standardizing text improves consistency, but it does not confirm whether names, locations, or email addresses are factually correct.

8. Interview Questions

Why is whitespace a data-quality issue?
Leading, trailing, or repeated spaces can make otherwise identical values appear different and may affect matching, filtering, and grouping.

What does TRIM do?
In Excel, TRIM removes leading and trailing spaces and reduces repeated spaces between words to a single space. In Python, str.strip() trims the ends, while split() and join() can also collapse repeated whitespace.

9. Conclusion

This task demonstrates a basic text-cleaning workflow using Python and Excel. The process standardizes common formatting issues and produces a cleaner dataset that is easier to work with in later analysis.

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